

ARFTR: Anchor-Restored Factorized Temporal Residual

Person-centered evidence fusion and a scenario-grouped component study

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Repository
<https://github.com/abdullahuseyinli-dot/arftr>

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Abstract

ARFTR combines an actor-memory anchor, complementary image/video probabilities and a template expert through posture and locomotion factors. A single same-track temporal pass adds neighboring evidence before decoding three activity classes. On 4,977 adaptively reused Okutama development centers, the retained system obtains 85.383648% macro-F1 and 85.895118% accuracy, with 702 errors. The matched anchor reaches 84.811129% macro-F1; ARFTR rescues 98 errors and harms 65 correct predictions. The +0.572519 percentage-point difference has a 95% scenario-bootstrap interval of $[-0.122986, +1.349570]$, so the component study does not establish statistically significant superiority. This report describes the retained computation, original seven-arm comparison and reproducibility boundary. It reports completed work, not a new evaluation or an independent confirmation result.

1. Task and contribution

The task is person-centered classification into sitting, standing, and combined walking/running. Evaluation uses supplied annotation-derived person boxes and track identities. It is not an end-to-end detection/tracking benchmark. The development population spans 11 Okutama scenarios, grouped into five outer folds; the same population was repeatedly used during architecture research.

The contribution is a constrained fusion layer: restore complementary evidence relative to a strong anchor, restrict one expert to the locomotion factor, and use track correspondence in a single temporal update. Frozen visual representations, actor-memory models and posture/motion factorization precede this layer; ARFTR does not claim a new pretrained backbone or the invention of factorization itself. The [architecture guide](#) links the upstream implementations and original protocol.

2. Factor-residual computation

Let M4 denote the selected actor-memory anchor, P6 the image/video/spatial consensus, and A3 the unrestricted-template expert. For each probability vector, posture log-odds distinguish sitting from upright activity; conditional-motion log-odds distinguish walking/running from standing:

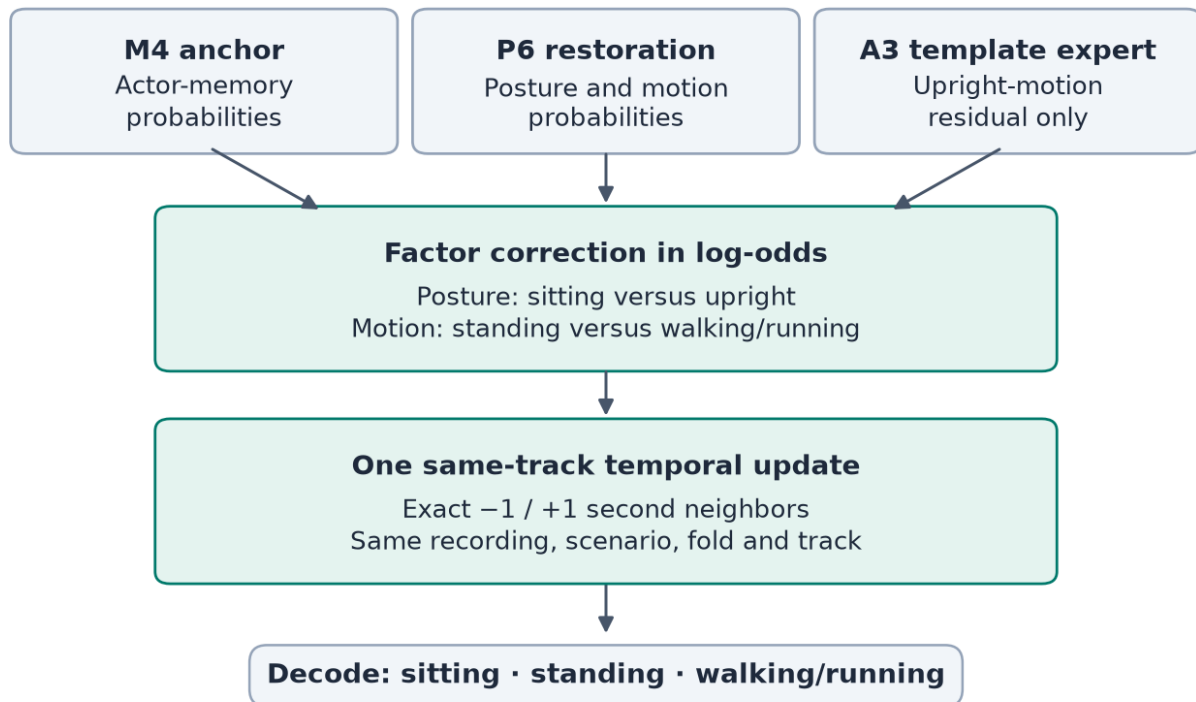
```
s = log(p_sitting / (p_standing + p_walking_running))
m = log(p_walking_running / p_standing)

s_residual = s_M4 + a * (s_P6 - s_M4)
m_residual = m_M4 + b * (m_P6 - m_M4) + c * (m_A3 - m_M4)
```

Input probabilities are clipped to $[1e-9, 1.0]$ before computing factors. For either factor z , one pass applies $z_{\text{final}} = z + \text{beta} * (\text{mean}(\text{valid_neighbor_}z) - z)$. Neighbors must be exactly -1 or $+1$ second away (30 frames), with matching scenario, outer fold, recording and track. A row with no valid neighbor receives no temporal update. Sitting probability is $\text{sigmoid}(s_{\text{final}})$; the remaining upright mass is split using $\text{sigmoid}(m_{\text{final}})$, with decoder logits clipped to $[-30, 30]$. All-zero coefficients take an explicit identity path, preserving the original M4 probability bytes.

ARFTR: anchored factor correction

Probability fusion above the upstream visual and actor-memory models



Four coefficients selected from inner out-of-fold predictions only.
All-zero correction retains M4 exactly. Offline / look-ahead inference.

Figure 1. Anchor-based factor correction and the same-track temporal pass.

3. Selection and component controls

Four coefficients are chosen from a fixed 300-candidate grid using scenario-held inner out-of-fold predictions inside each outer training fold. Selection orders candidates by macro-F1, NLL, coefficient sum, then lexicographic coefficients. The selected transformation is applied separately to aligned upstream seeds 42, 43 and 44; final probabilities are their arithmetic mean. ARFTR adds no new neural fits, but this does not make upstream feature extraction or training free.

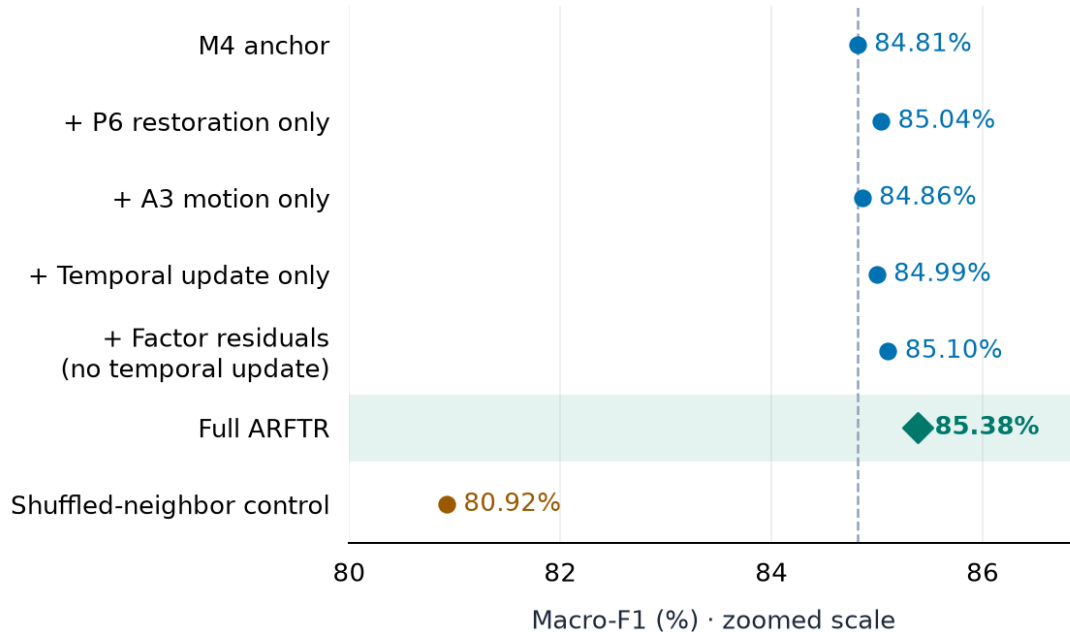
Seven original arms compare the exact M4 anchor, P6 restoration only, A3 motion residual only, temporal update only, factor residuals without the temporal pass, full ARFTR, and shuffled neighbors. Component-removal arms reuse the full model's selected coefficients; they are not separately retuned. The shuffled control preserves each row's neighbor count and samples within the same scenario, excluding self and true neighbors. It tests correspondence in this recipe, not pure motion independently of appearance.

The [locked protocol](#) specifies selection, the seed aggregation, original success criteria and statistical tests. Activity labels and diagnostic support categories do not enter the inference functions. Future-frame evidence is permitted, making the evaluated system offline or look-ahead rather than a validated causal streaming model. Upstream context can extend beyond this layer's ± 1 -second neighborhood.

4. Original component results

ARFTR: original component evidence

4,977 centers / 5 grouped folds / 3-seed predictions



Points: observed scores. Dashed line: M4 anchor.
Removal arms reuse the full model's selected coefficients.

Full – anchor: +0.57 pp; 95% interval [-0.12, +1.35] pp.

Figure 2. Original seven-arm component study on the same development population.

Matched system	Macro-F1	Accuracy	Errors	NLL	Brier sum
Exact M4 anchor	84.811129%	85.232068%	735	0.390042	0.220730
Full ARFTR	85.383648%	85.895118%	702	0.377349	0.213505

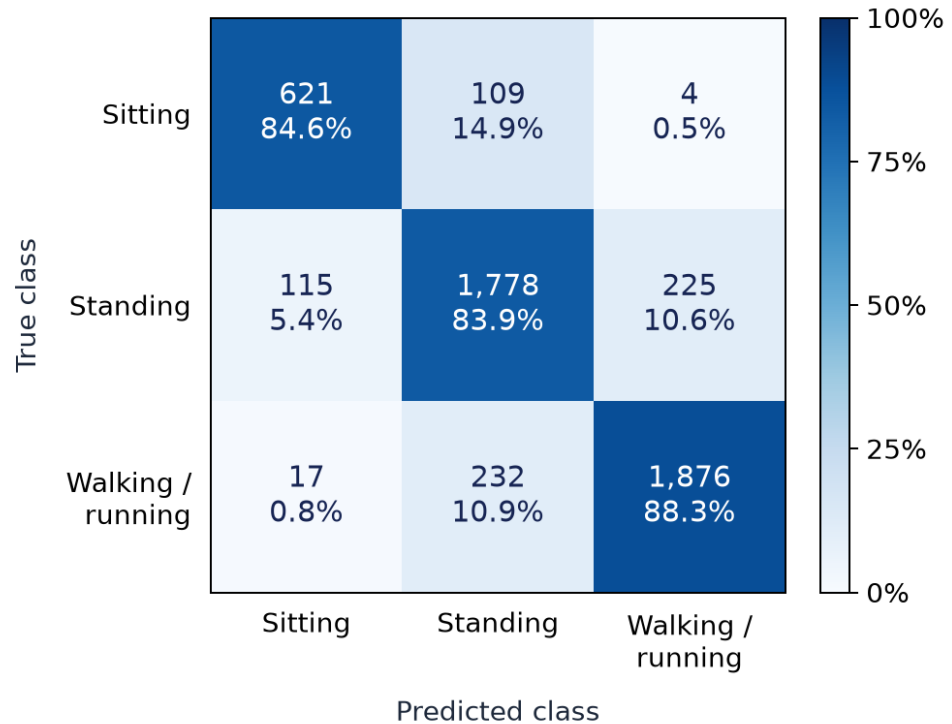
The complete arm metrics, confusion matrices and statistical exports are available in the [component evidence](#). Full ARFTR makes 33 net corrections relative to M4: 98 rescues and 65 harms. The marginal component scores should not be added to explain the full-model gain.

The paired macro-F1 interval uses 10,000 bootstrap resamples of the 11 scenarios. Its [-0.122986, +1.349570] percentage-point range includes zero. The exact one-sided scenario-swap test enumerates all 2,048 assignments and yields $p = 0.064941$, above the original 0.05 threshold. The strict original screen did not pass in full. Retention is a documented development decision, not established superiority or an external state-of-the-art claim. Neither the interval nor scenario-grouped cross-fitting removes adaptive-selection bias from repeated research use.

5. Error structure and development lineage

Retained ARFTR: 702 remaining errors

Cells show counts and percentages within each true class



Standing vs walking/running: 457 of 702 errors (65.1%).

Adaptive development; error concentration is not evidence of a cause.

Figure 3. Retained ARFTR confusion matrix: true classes in rows, predictions in columns.

Class	Examples	Precision	Recall	F1
Sitting	734	82.47%	84.60%	83.52%
Standing	2,118	83.91%	83.95%	83.93%
Walking / running	2,125	89.12%	88.28%	88.70%

Standing/walking-running confusion accounts for 457 of 702 errors (65.1%). This locates the dominant error pair; it does not prove a motion-feature failure or establish that all such examples are visually resolvable. Camera movement, small actors, occlusion, pose ambiguity and label semantics remain possible explanations.

The earliest T2 system reached 71.923768% macro-F1 with 1,412 errors on the same centers. The historical increase to ARFTR is +13.459880 points and 710 fewer errors. It combines representation, source, memory and fusion changes; only the narrower M4 comparison isolates the addition of the ARFTR layer to the matched anchor. Earlier sealed temporal confirmation used different models and 1,771 separate examples; its 78.54% score is not an ARFTR confirmation result.

Later correction and sequence trials did not establish a replacement. Some exploratory point estimates were higher but inconsistent across folds. The final matched paired-null motion residual reached 85.311396%, failed its fixed gates, and was not promoted. The [research history](#) preserves those outcomes; they do not imply a universal ceiling for future methods.

6. Reproducibility and scope

The public package provides code, locked protocols, aggregate component metrics, confusion matrices, a selected 16-result ledger and a curated knowledge graph. It does not redistribute dataset media, model weights or private feature caches. The aggregate checker needs only Python's standard library:

```
python tools/check_project.py
```

It recomputes confusion-based macro-F1, accuracy, per-class F1, errors and net corrections, and checks evidence inventories, hashes, model-retention gates and current presentation assets. NLL, Brier, rescue/harm overlap and scenario-based intervals require original row predictions for full numerical replay; verifying their export hashes is not equivalent to recomputing them.

On the research workstation, a separate preservation check covers protected research files, original document snapshots, the final experiment and its locked dependencies. Full model replay additionally requires compatible environments, upstream models, data and ancestry-safe caches. Unit tests, public arithmetic checks and saved-output replay are distinct from rerunning all upstream neural models. The [reproduction guide](#) and [validation record](#) state the checks performed and their limits.

7. Interpretation

The useful mechanism is explicit, anchor-relative control of complementary evidence: P6 can restore both factors, A3 can alter only conditional locomotion, and temporal correspondence is constrained before averaging. The shuffled-neighbor result and matched rescue/harm evidence support this design as a development finding, with the uncertainty stated above. They do not establish causality for each historical gain, robustness to predicted tracks, or deployment performance.

The retained 85.38% development result is preserved. Stronger generalization claims would require a frozen recipe and untouched scenarios or an external population. No clinical, demographic-fairness or consequential individual-decision validation has been performed. The [model card](#) and [third-party notices](#) document intended use and data/model terms.

Evidence and implementation

- [ARFTR implementation and evaluation runner](#).
- [Original component export and provenance](#).
- [Retained metrics and selected experimental lineage](#).
- [Earlier development review](#).

The current report summarizes existing evidence. Historical reports, numerical exports and experiment protocols remain unchanged.